

Event-Causal RAG: A Retrieval-Augmented Generation Framework for Long Video Reasoning in Complex Scenarios

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Abstract

Recent large vision-language models have achieved strong performance on short- and medium-length video understanding, yet they remain inadequate for ultra-long or even infinite video reasoning, where models must preserve coherent memory over extended durations and infer causal dependencies across temporally distant events. Existing end-to-end video understanding methods are fundamentally limited by the $O(n^2)$ complexity of self-attention, while recent retrieval-augmented generation (RAG) approaches still suffer from fragmented clip-level memory, weak modeling of temporal and causal structure, and high storage and online inference costs. We present Event-Causal RAG, a lightweight retrieval-augmented framework for infinite long-video reasoning. Instead of indexing fixed-length clips, our method segments streaming videos into semantically coherent events and represents each event as a structured State-Event-State (SES) graph, capturing the event together with its surrounding state transitions. These graphs are merged into a global Event Knowledge Graph and stored in a dual-store memory that supports both semantic matching and causal-topological retrieval. On top of this memory, we design a bidirectional retrieval strategy to efficiently identify the most relevant event causal chains and provide them, together with the associated video evidence, to a backbone video foundation model for answer generation. Experiments on long-video understanding benchmarks demonstrate that Event-Causal RAG consistently outperforms strong clip-based retrieval baselines and long-context video models,

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particularly on questions requiring multi-event integration and causal inference across long temporal gaps, while also achieving improved memory efficiency and robust streaming performance.

1 Introduction

Recent advances in large vision-language models (LVLMs) have substantially improved the performance of video understanding systems [1, 2, 3, 4, 5]. In particular, mainstream video foundation models have become increasingly capable of recognizing actions, summarizing content, and answering questions over short to medium-length videos [1, 2, 6, 3]. These successes, however, do not directly translate to ultra-long or even infinite video reasoning. Unlike short-form video understanding, long-horizon reasoning requires a model to track evolving storylines, maintain coherent memory over extended time spans, and infer causal dependencies among events that may be separated by minutes or even hours [7, 8, 9, 10, 11]. Such requirements are central to real-world scenarios including surveillance [12, 13], medical monitoring, and traffic analysis [14, 15], where decisions often depend not only on *what* happened, but also on *why* it happened and how it unfolded over time.

Current end-to-end VLM-based video understanding solutions face a severe computational bottleneck. The dominant Transformer architecture relies on self-attention, whose time and memory complexity increase rapidly with the number of visual tokens [16]. As video duration grows, the tokenized representation quickly becomes prohibitively large, leading to excessive latency, memory overflow, and unstable inference. Although recent efforts have explored temporal pooling, token compression, sparse attention, and context window extension to alleviate this issue [17, 18, 19, 20, 21, 22, 23, 24], these techniques do not fundamentally resolve the challenge of reasoning over extremely long sequences. In practice, models still suffer from the well-known lost-in-the-middle phenomenon [25], where information located far from the beginning or end of the context is poorly utilized. As a result, simply extending the context window does not yield reliable long-horizon reasoning; instead, it often increases computational cost while degrading accuracy on causal and temporally compositional queries [26, 27, 11].

To avoid processing an entire long video at once, recent works have adopted retrieval-augmented generation (RAG) for long-video understanding [28, 29, 30, 31, 32, 33]. Nevertheless, existing video-RAG approaches still exhibit three critical limitations: **(1)** They often **fragment spatiotemporal context** by constructing memory over fixed-duration clips rather than semantically complete events. In most existing pipelines, a long video is mechanically segmented into windows of equal length, e.g., every 5 or 10 seconds, and each segment is independently embedded. While convenient, this design ignores the natural boundaries of events. A physically and semantically coherent causal chain may therefore be split across multiple clips [10, 34, 27]. **(2)** Existing retrieval strategies primarily rely on **global semantic similarity** between the query and individual clips, while largely ignoring explicit modeling of temporal dependency and causal relations between events. This design is sufficient when the goal is to find a visually similar scene or a directly matching clip, but it is inadequate for questions that require multi-step reasoning across distant observations [35, 7, 36, 37]. **(3)** Current RAG-based methods **remain costly in both storage and online processing**. Because they store dense representations for large numbers of low-level clips, the memory overhead grows rapidly with video duration. This issue becomes more severe in streaming environments, where the system must continuously process and retrieve new content as the video arrives [38, 39, 40, 41, 42]. Meanwhile, large-model inference is often too slow to keep pace with high-throughput streams, forcing practitioners to rely on expensive parallel deployment for near-real-time performance.

In this paper, we propose Event-Causal RAG, a retrieval-augmented framework for INFINITE long-video reasoning. Our key idea is to replace clip-level memory with a State-Event-State (SES) graph memory that aligns retrieval units with semantically complete events rather than arbitrary temporal windows. Specifically, we first perform asynchronous event-driven segmentation on streaming videos using joint visual and audio cues, so that each memory unit preserves a naturally coherent event whenever possible. Next, we parse each clip into structured event graphs, which capture the core events with their pre- and post-event states. The event graphs are unified into a global Event Knowledge Graph and stored in a dual-store memory (a Vector DB for node semantics and a Graph DB for topology). Compared with conventional vector databases that mainly support similarity retrieval, our memory explicitly retains higher-level relational information, allowing the system to retrieve not only semantically relevant events but also causally connected context. Finally, we

use a bidirectional retrieval algorithm for RAG that can rapidly find the most relevant event causal chains. The retrieved event graphs are serialized and provided, together with the corresponding event clips, to the backbone foundation VLM, thereby enhancing its event-causal reasoning capability and enabling the generation of detailed responses. Importantly, Event-Causal RAG is lightweight and can be integrated with existing video foundation models without expensive end-to-end retraining over ultra-long sequences.

We evaluate our approach on long-video understanding benchmarks, and experimental results show that Event-Causal RAG consistently outperforms strong clip-based retrieval baselines and long-context video models, especially on questions that require multi-event integration and causal inference across extended temporal gaps. Moreover, our method achieves favorable efficiency by reducing memory footprint while maintaining robust performance in streaming scenarios. In summary, EC-RAG addresses the three core challenges identified above: **(a)** It mitigates spatiotemporal fragmentation by constructing event-level memory, **(b)** strengthens long-horizon reasoning through explicit State-Event-State causal structure, and **(c)** reduces storage and online inference overhead through compact dual-store retrieval. These properties make EC-RAG suitable for ultra-long and streaming video scenarios where robust reasoning requires both efficient memory management and temporally grounded causal evidence.

2 Related Work

Long Video Understanding via LVLMs. Recent large vision-language models (LVLMs) have substantially advanced video understanding, enabling action recognition, temporal description, and question answering over short and medium-length inputs [1, 2, 3, 6]. To scale LVLMs to longer videos, existing studies explore frame sampling, temporal pooling, visual token compression, context-window extension, positional encoding adaptation, KV-cache compression, and streaming inference [17, 18, 19, 20, 21, 38, 39, 5]. These methods improve the computational feasibility of processing long visual sequences. However, merely extending the accessible context does not resolve the core challenge of long-video understanding, since the context window of LLMs is ultimately bounded and cannot scale to ultra-long video inputs. Retrieval-augmented generation (RAG) with external memory offer an alternative by decoupling video processing from question answering [2, 28, 29, 30]. Recent scene-level and graph-based RAG variants further show that structured memory can improve multi-hop retrieval and long-range reasoning [31, 43, 44, 45]. Nevertheless, most existing RAG pipelines still construct memory over fixed-duration clips, coarse scenes, or generic semantic graphs. Such memory units can break semantically complete events, while embedding-based retrieval mainly captures query similarity and lacks explicit modeling of directional state changes and causal chains.

Temporal-causal Reasoning in Long Videos. Prior research on video understanding has extensively studied temporal reasoning, including event ordering, temporal localization, and long-range dependency modeling [35, 10, 7, 9, 8, 27, 11]. Early approaches mainly focus on short clips or temporally localized events, where temporal dynamics can be captured within limited contexts. To handle longer videos, subsequent works explore hierarchical representations, memory-based architectures, and retrieval-based frameworks to improve the modeling of extended temporal structure [7, 9, 8]. Meanwhile, another line of work investigates causal reasoning in videos, such as event causality modeling, causal temporal grounding, and counterfactual analysis [10, 27, 46, 47, 11], aiming to move beyond temporal correlation toward a deeper understanding of event relations. However, these methods do not effectively capture long-range causal dependencies in extremely long videos. Unlike prior approaches, EC-RAG represents streaming videos as a global Event Knowledge Graph. This design explicitly preserves overall pre-event and post-event state transitions, and supports long causal-topological retrieval, thereby facilitating ultra-long or streaming video reasoning.

3 Methodology

The overview of the EC-RAG framework is illustrated in Figure 1, consisting of three modules. First, the Dual-Sentinel Event Segmentation module (Section 3.1) cuts the input videos into full-event clips. Next, the SES Event Memory Constructor (Section 3.2) extracts the SES graphs and stores the

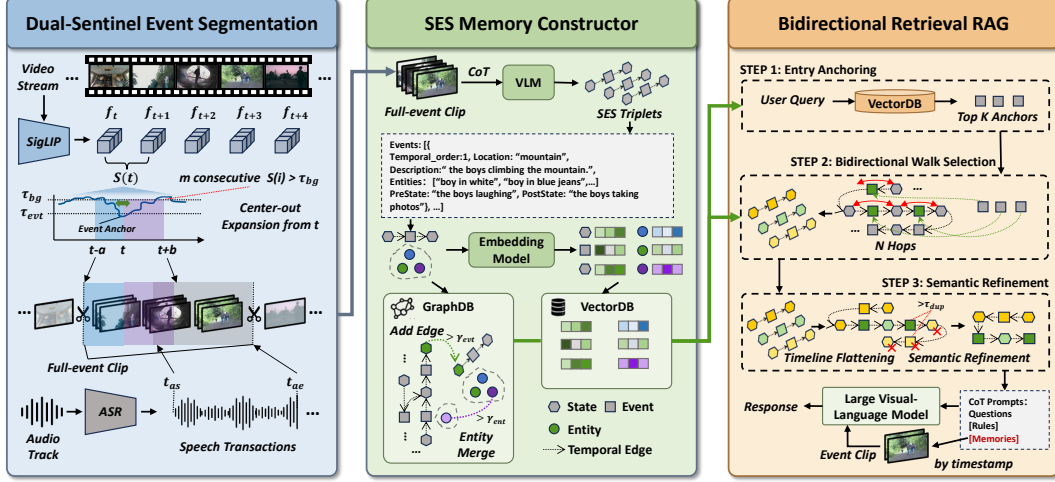


Figure 1: Overview of the Event-Causal RAG framework.

topology and node representations. With the event memory, the RAG module (Section 3.3) can query event causal chains and enhance LVL generation.

3.1 Dual-Sentinel Event Segmentation

Visual Sentinel. The foundational step of EC-RAG is determining where an event begins and ends. Let a video stream be $V = \{f_1, f_2, \dots, f_T\}$, where f_i is a visual frame. We first utilize a pre-trained SigLIP [48] visual encoder E_v to extract frame-level embeddings and then calculate the similarity between adjacent frames:

$$S(t) = \cos(E_v(f_t), E_v(f_{t+1})), \quad (1)$$

where $\cos(\cdot, \cdot)$ denotes cosine similarity. We then define a trigger to anchor an event by detecting fluctuations in $S(t)$. When $S(t)$ falls below a highly sensitive threshold ($\tau_{evt} = 0.97$), it signifies a micro-action or state transition. To prevent over-segmentation from high-frequency noise, we add a local minima constraint. Thus, an event is anchored at t if:

$$S(t) < \tau_{evt} \wedge S(t) < S(t-1) \wedge S(t) < S(t+1). \quad (2)$$

From these event anchors, we employ a “center-out” expansion until a stable background threshold ($\tau_{bg} = 0.99$) is exceeded to confirm an event window $T(t) = \{f_{t-a}, \dots, f_t, \dots, f_{t+b}\}$:

$$S(e)|_{t-a < e < t+b} < \tau_{bg} \wedge S(t-a-1) > \tau_{bg} \wedge S(t+b) > \tau_{bg}. \quad (3)$$

To avoid overly fine-grained or excessively long clip segmentation, we also set lower and upper bounds (p_l, p_u) on the event window length: $p_l < a + b < p_u$. Finally, by applying the segmentation algorithm, we obtain video chunks containing complete events.

Audio Sentinel. Simultaneously, to capture audio events that are not reflected in the visual signal, we adopt an off-the-shelf ASR tool to extract video speech, projecting transcriptions (with strict punctuation preservation) directly into the corresponding temporal segments. This cross-modal projection preserves the emotional and tonal context crucial for downstream LLM reasoning. The result is a strict, seamless temporal partition of Dynamic Events (sampled at high FPS) and Static Backgrounds (sampled at low FPS). The detailed algorithm is available in Appendix 1.

3.2 State-Event-State Memory Constructor

Entity-First SES Triplets Extraction. We prompt a VLM to extract (State-Event-State) SES triplets from each event clip $T(t)$. Specifically, we use a two-step CoT guidance, where the VLM is explicitly constrained to first output a `scene_inventory`, enumerating all visible interacting entities (e.g., ["red-shirt man", "glass door", "military helicopter"]). This step acts as a

spatial attention dispersal mechanism, forcing the VLM to track each listed entity independently. Subsequently, the VLM generates structured SES triplets in the form of:

$$State_{pre} \xrightarrow{[:PRECONDITION_OF]} Event \xrightarrow{[:RESULTS_IN]} State_{post} \quad (4)$$

Each event node contains the natural-language description of the event, the location, and the involved entities, while each state node describes the pre- and post-event states. Additionally, an explicit `temporal_order` attribute is assigned to each event triplet to enforce chronological logic and prevent causal hallucination.

Dual-Store Memory System. We utilize a Dual-Store Memory system comprising a Vector DB (e.g. Milvus) for high-dimensional semantic routing and a Graph DB (e.g. Neo4j) for topological mapping. The extracted triplets from individual event clips must be continuously stored and unified with previous ones. We employ an embedding model (e.g., Qwen3-Embedding-4B) to compute node representations. First, for each newly arrived SES triplet, if its entity similarity with an existing node exceeds $\gamma_{ent} = 0.85$, the two are merged into a single global node. We then enforce **State Continuity** to construct causal chains: if the *PostState* of chunk $Event_i$ and the *PreState* of chunk $Event_j$ exhibit a semantic similarity $\gamma_{evt} = 0.85$, they are fused, and a `[:TEMPORAL_NEXT]` edge is established, weaving parallel action chains into a dense, interconnected mesh.

3.3 Bidirectional Retrieval RAG

During the RAG QA phase, standard path-finding algorithms suffer from exponential traversal explosion in dense graphs. We mitigate this via a three-step dimension-reduction algorithm:

Step 1: Entry Anchoring. We query the vector database to retrieve the most semantically relevant event nodes as entry points.

Step 2: Bidirectional Walk Selection. Within the graph database, we abandon directed path enumeration. Instead, we collect all distinct nodes within N hops (typically $N = 2$) of the anchors, capturing both causes and effects without path duplication.

Step 3: Semantic Refinement. To present the LLM with a clean context, all retrieved nodes are flattened chronologically based on physical timestamps. To eliminate the ‘‘Context Redundancy’’ caused by VLMs repeatedly extracting identical continuous actions across adjacent chunks, we introduce an embedding-based semantic deduplication mechanism. Let n_i denote a newly retrieved candidate node, and let $n_i.\text{Text}$ represent its extracted textual description. We utilize the embedding model Ψ_e to project this text into a semantic vector $\Psi_e(n_i.\text{Text})$. We then compute its maximum cosine similarity, $S_{max}(n_i)$, against $\mathcal{M}_{seen}$, which is the set of embeddings for all previously preserved nodes:

$$S_{max}(n_i) = \max_{v \in \mathcal{M}_{seen}} \{\cos(\Psi_e(n_i.\text{Text}), v)\}, \quad (5)$$

where $\cos(\cdot, \cdot)$ denotes the cosine similarity function. If $S_{max}(n_i) > \tau_{dup}$, where τ_{dup} is a predefined deduplication threshold, the candidate node is considered redundant and is pruned. Otherwise, its semantic vector $\Psi_e(n_i.\text{Text})$ is appended to $\mathcal{M}_{seen}$. This semantic state collapse automatically compresses prolonged, unchanging scenes into a single script entry, reducing the context volume by $> 50\%$ and drastically increasing information density. Finally, the LLM executes Chain-of-Thought (CoT) reasoning based on this purified prompt. The relevant event clips are also provided as visual inputs.

4 Experiments and Analysis

4.1 Settings

Multi-scale Video Benchmarks. To comprehensively verify the system’s robustness across different temporal spans, we constructed a hierarchical evaluation framework ranging from seconds to a full day: 1) **NExT-QA** (short videos regarding the causal and temporal relationships between actions, 5-180s); 2) **EventBench** (medium-to-long videos, multi-level event understanding and causal reasoning, 60-1800s); 3) **Video-MME Long** (extremely long videos, including two chapters: event action understanding and action reasoning, 1800-3600s); 4) **24-hour+ streaming surveillance video** (the video uses real-world surveillance footage with a time span of over a day to verify the framework’s ability to process unlimited streams of videos).

Table 1: **Video-MME Long (1800s-3600s) subset testing.** Under extreme length challenges, EC-RAG achieves substantial improvements (+12.50%) in capturing dynamic features like action recognition through semantic dimensionality reduction and graph abstraction.

Task	Qwen3-VL-8B(Baseline)	EC-RAG (Ours)	Δ (Improvement)
Action Reasoning	43.89	46.67	+ 2.78
Action Recognition	35.00	47.50	+ 12.50

Table 2: **Quantitative evaluation on EventBench and NExT-QA.** EC-RAG improves all open-source backbones on NExT-QA and overall EventBench accuracy, with the main gains coming from contextual and episodic reasoning. Across overall metrics, EC-RAG enables all three open-source backbones to match or surpass the GPT-4o overall score on EventBench.

Method	NExT-QA (5–180s)	EventBench (60–1800s)			
	Accuracy (%)	Contextual	Episodic	Counter-intuitive	Overall
GPT-4o (Closed-source)	76.71	50.13	37.33	63.44	53.33
Gemini1.5-pro (Closed-source)	75.74	32.15	38.67	52.86	43.24
Qwen3-VL-8B	73.08	53.16	45.67	59.91	52.39
+EC-RAG	75.54 (+2.46)	61.77 (+8.61)	48.33 (+2.66)	53.30	55.31 (+2.92)
VideoLlama3-7B	76.69	56.71	42.00	55.95	51.74
+EC-RAG	78.48 (+1.79)	61.01 (+4.30)	49.67 (+7.67)	55.51	55.97 (+4.23)
InternVL3.5-8B	77.54	56.96	45.00	60.35	53.90
+EC-RAG	80.54 (+3.00)	62.28 (+5.32)	46.67 (+1.67)	57.27	55.97 (+2.07)

Implementation Details. EC-RAG utilizes SigLIP [48] for visual encoding, adaptively sampling dynamic events at 8 FPS and static backgrounds at 2 FPS to optimize computational overhead. The Dual-Modal Sentinel strictly bounds streaming chunks to 12 seconds to prevent context overflow. Within the dual-store memory, nodes are embedded via Qwen3-Embedding-4B into ChromaDB for semantic routing, while causal topologies are preserved in Neo4j. Bidirectional traversal is constrained to a maximum depth of $N = 2$, utilizing a semantic deduplication threshold of $\tau_{dup} = 0.85$. All evaluations, including the 24-hour infinite stream test, are executed entirely on a single NVIDIA RTX 5090 (32GB VRAM).

4.2 Verification of causal reasoning ability under ultra-long videos (1800s-3600s)

In the extremely long video subset of Video-MME (Long), which lasts up to 1 hour, native large models face severe context window constraints. As shown in Table 1, EC-RAG demonstrates significant advantages. In the Action Reasoning task, EC-RAG achieves a score of 46.67%, representing a +2.78% improvement over the native end-to-end no-RAG baseline (43.89%).

Notably, in the Action Recognition task, EC-RAG increases the accuracy from 35.00% to **47.50%**, achieving a substantial growth of **+12.50%**. This indicates that when the video length exceeds 1800s, native models are prone to missing key dynamics due to attention dispersion. In contrast, EC-RAG, based on the dual-sentinel diffusion chunking and EKG state transition abstraction, effectively localizes the causal anchors of actions, thereby significantly enhancing dynamic perception capabilities in extremely long sequences.

4.3 Causal reasoning ability within the framework of short and medium length videos

Robustness in Short-Horizon Micro-Dynamics. As shown in Table 2, on the NExT-QA benchmark (5–180s), EC-RAG consistently improves accuracy across all evaluated open-source backbones, with gains ranging from +1.79% to +3.00%. These consistent gains indicate that the proposed State-Event-State (SES) abstraction does not only benefit long videos, but also strengthens short-horizon micro-dynamic reasoning by preserving fine-grained spatiotemporal evidence while filtering visually redundant context. This is important because short-video QA still often requires identifying subtle action transitions and local temporal dependencies, which can be weakened by naive text-only filtering or coarse clip summarization.

Efficacy in Long-Horizon Causal Reasoning. The advantage of EC-RAG becomes more evident on EventBench (60–1800s), where questions require integrating temporally distributed evidence. In the

Contextual reasoning split, EC-RAG improves all three open-source models, with gains ranging from +4.30% to +8.61%. These results show that dual-store state merging and bidirectional graph retrieval help recover contextual evidence that is distributed across multiple events rather than concentrated in a single clip. A similar trend appears in the *Episodic* split, where EC-RAG improves all three backbones with gains ranging from +1.67% to +7.67%. The particularly large episodic gain for VideoLlama3-7B suggests that event-level topological memory can mitigate feature amnesia and logical fragmentation when the backbone model alone struggles to preserve long-range event continuity.

Performance Variance in Subjective Queries. While EC-RAG consistently improves objective and temporally grounded reasoning, it exhibits a different trend on the *Counter-intuitive* subset. This subset contains more subjective, aesthetic, or abstract questions (e.g., “*What makes this video so magical?*”), for which the answer often depends on implicit visual impression rather than explicit physical state transitions. Because EC-RAG converts videos into fact-grounded SES structures, it prioritizes observable entities, actions, and state changes, but may suppress ambiguous artistic cues or high-level affective semantics that are useful for such subjective interpretation. This explains why EC-RAG can slightly reduce performance on the *Counter-intuitive* split while still improving the overall reasoning capability.

Importantly, the overall results remain strongly aligned with the intended design of EC-RAG. On EventBench, EC-RAG improves overall accuracy across all three backbones, with gains ranging from +2.07% to +4.23%. Thus, all three parameter-efficient open-source models match or surpass the GPT-4o overall score of 53.33%, while relying on compact event-causal memory rather than full-context video ingestion or proprietary model scale. These results support the central claim of this paper: structured event-level memory and causal-topological retrieval provide a practical mechanism for improving long-video reasoning, especially when answers require multi-event integration and temporally grounded causal evidence.

4.4 Ablation Study

To verify the systematic necessity of each core component of EC-RAG, we conducted an in-depth module ablation on EventBench (see Table 3). To explicitly eliminate the performance variance introduced by raw video frame parsing and strictly isolate the structural effectiveness of the framework itself, this ablation study adopts a “pure RAG with text-based reasoning” paradigm. Under this controlled setting, the complete EC-RAG architecture utilizing Qwen3-VL-8B establishes a strong baseline accuracy of **49.37%**.

The experimental data reveals a clear logic: lacking the elastic sentinel with context redundancy wrapper (degrading to a fixed chunking with mechanical truncation) severs the integrity of underlying actions, leading to a 2.53% decline. Removing the local SES abstraction based on event transitions causes the model to lose its dynamic causal focus, resulting in a noticeable drop of **6.33%**. Failing to perform dual-store topological merging and relying solely on isolated chunks completely breaks long-range logic, causing a 10.08% decrease. Finally, removing the lightweight semantic filtering deduplication causes graph traversal redundancy due to synonymous static descriptions, which dilutes the model’s attention and leads to a 3.04% degradation.

Table 3: **System Ablation Study on EventBench (Contextual).** Each component provides significant and indispensable performance gains in architecture integration, macroscopic causal reconstruction, and redundancy suppression.

Model Variant	Accuracy (%)
EC-RAG (Full System Architecture)	49.37
<i>w/o Elastic bidirectional diffusion sentinel (degrades to fixed chunks without perception detection)</i>	46.84 (-2.53)
<i>w/o Local SES abstraction (degrades to unstructured text summaries without specific dynamic structures)</i>	43.04 (-6.33)
<i>w/o Dual-store topological state merging (degrades to single-hop blind matching with pure dense vectors [29])</i>	38.99 (-10.08)
<i>w/o Semantic embedding filtering deduplication (degrades to redundant direct information acquired solely by hop count)</i>	46.33 (-3.04)

4.5 24-Hour Infinite Stream Processing: Industrial-grade Quantitative Tracing Test

To thoroughly evaluate the framework’s deployment capability on extremely long real-world data, we fed the system a continuous 24-hour unedited industrial security surveillance stream.

We extracted the independent event segments automatically chunked by the perception layer, input them into a high-performance closed-source large model (Qwen3.5), and conducted rigorous blind testing combined with human review. The evaluation adopts "Strict Accuracy," requiring the extraction to simultaneously satisfy: (1) no hallucination in semantic actions; (2) the time window covers the complete event cycle; (3) the entity feature descriptions match; (4) the location is correct.

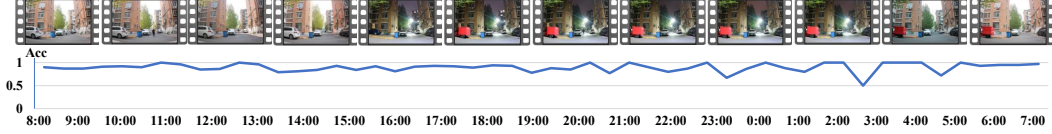


Figure 2: The hourly strict accuracy rate of the 24-hour security surveillance stream.

Utilizing a single RTX 5090 with 32GB of VRAM, the framework continuously processed the 24-hour video. The system adaptively generated 2514 distinguishable event chunk samples, taking a total of 11 hours and 14 minutes, fully meeting the requirements for streaming processing. Among these, 2277 events fully met the four-dimensional strict matching criteria. Under the condition of using only an 8B base model for cognitive extraction, EC-RAG achieved a **strict information extraction accuracy of 90.57%**, and the hourly accuracy is shown in Figure 2. This large-scale quantitative data confirms that by leveraging the diffusion sentinel and state-merged extraction, EC-RAG not only breaks the physical constraints of extreme contexts but also maintains highly stable event recognition over extensive spatiotemporal spans. By using only a single RTX 5090 and an 8B-scale VLM, high-precision video event causal graph extraction can be achieved.

4.6 Qualitative Analysis: Graph Construction and Dual-Store Retrieval

Figure 3 details the EC-RAG pipeline for processing a cooking instructional video, illustrating both Event Knowledge Graph (EKG) construction and dual-store retrieval.

SES Graph Construction (Figure 3a). The left panel depicts spatiotemporal discretization into a State-Event-State (SES) chain. Instead of encoding redundant frames, EC-RAG abstracts critical physical transformations (e.g., S_0 : intact shell $\xrightarrow{E_1: \text{crack egg}}$ S_1 : liquid yolk). Linking these tuples via `[ :TEMPORAL_NEXT ]` relations based on post-state similarities successfully compresses the continuous video into a discrete causal graph.

Dual-Store Retrieval and Generation (Figure 3b). The right panel demonstrates dual-store retrieval for the query "How did the chef make custard sauce?". Pure vector retrieval yields only fragmented semantic anchors (A_1 : custard, A_2 : eggs, A_3 : cream), inevitably causing standard RAG systems to hallucinate. Conversely, EC-RAG uses these anchors to initiate a bidirectional graph traversal (max depth = 2), recovering missing intermediate causal steps (E_1, E_2, E_4). Following temporal merging and semantic deduplication, the compiled chronological prompt enables the LLM to generate a precise, hallucination-free answer.

4.7 VRAM Consumption and Infinite Stream Processing

To evaluate the system-level efficiency and hardware requirements of processing long-form videos, we analyze the Video RAM (VRAM) consumption of an 8B-parameter VLM under both the native unoptimized setting and the EC-RAG framework.

As illustrated in Figure 4(a), processing continuous video tokens natively poses a severe memory bottleneck. Even with Grouped-Query Attention (GQA) optimizations (8 KV Heads, ~ 0.000122 GB/Token), the KV-cache accumulates linearly with the input video length. Consequently, the native VLM rapidly exhausts the standard 32GB hardware limit, reaching a critical Out-Of-Memory (OOM) point at approximately **162 seconds**. This hardware wall fundamentally prohibits standard VLMs from executing medium-to-long video reasoning tasks.

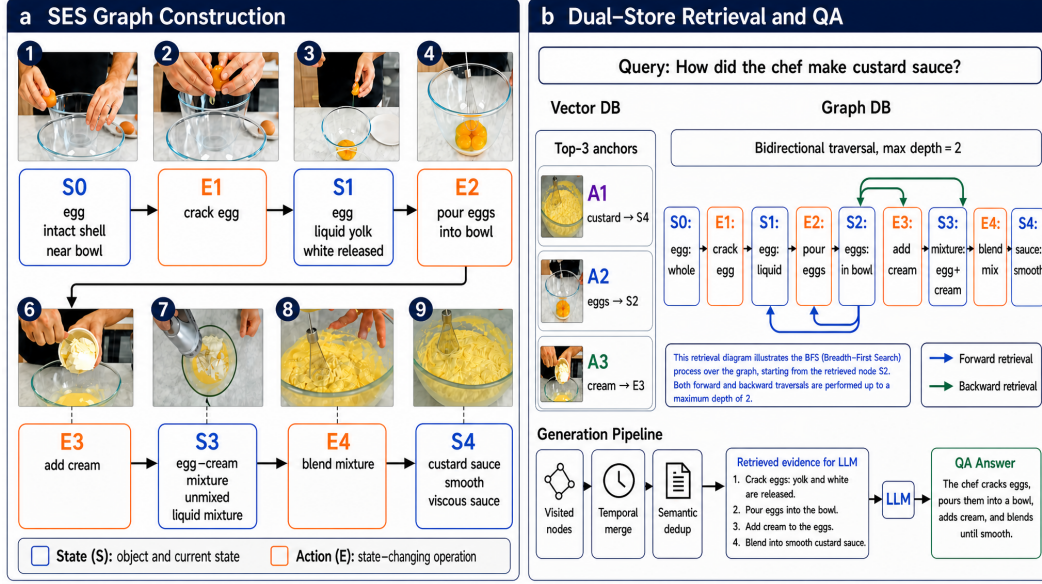
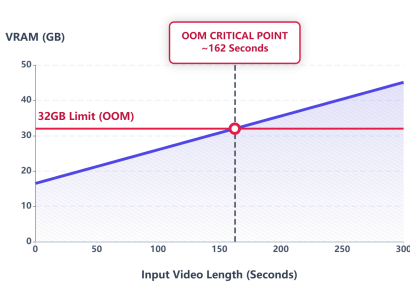


Figure 3: **Qualitative pipeline of EC-RAG.** (a) **SES Graph Construction:** Continuous video frames are parsed into alternating nodes of entity States (S) and Actions (E), compressing pixels into a discrete causal chain. (b) **Dual-Store Retrieval and QA:** For a complex causal query, pure vector retrieval only locates isolated semantic anchors (A_1, A_2, A_3). EC-RAG utilizes these anchors to trigger a bidirectional graph traversal, reconstructing the complete chronological evidence chain required for accurate LLM generation.

In stark contrast, Figure 4(b) demonstrates the bounded memory footprint achieved by EC-RAG. By implementing an elastic streaming architecture with a maximum chunking window of 12 seconds, the framework entirely decouples VRAM consumption from the total video duration. The peak memory utilization strictly plateaus at ~ 17.6 GB. Once a 12-second visual chunk is processed and abstracted into a discrete State-Event-State (SES) textual node, the heavy visual KV-cache is flushed, and the semantic state is persistently stored in the Graph DB.

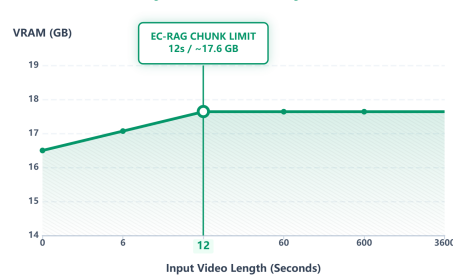
This architectural paradigm successfully prevents OOM crashes, enabling 8B-scale open-source models to process infinitely long video streams (e.g., 3600 seconds and beyond) with zero memory degradation, making it highly deployable on standard consumer-grade GPUs.

Unoptimized GQA VRAM Consumption (8B VLM)



(a) Unoptimized GQA VRAM Consumption

EC-RAG Framework VRAM Consumption (8B VLM)



(b) EC-RAG VRAM Consumption

Figure 4: **VRAM consumption comparison during video processing.** (a) The native 8B VLM accumulates KV-cache linearly, hitting a 32GB OOM wall at ~ 162 seconds. (b) EC-RAG, bounded by a 12s maximum chunking strategy, plateaus at ~ 17.6 GB, enabling infinite stream processing without memory explosion.

5 Conclusion and Limitations

In this paper, we introduced EC-RAG, a lightweight retrieval-augmented framework for infinite long-video reasoning. By replacing fixed clip-level memory with a graph memory, EC-RAG preserves semantically coherent events and explicitly models temporal and causal dependencies across long horizons. Built upon asynchronous event-driven segmentation, a compact dual-store event memory, and bidirectional retrieval, our framework enables video foundation models to retrieve not only semantically relevant content but also causally connected context, without expensive end-to-end retraining over ultra-long sequences. Experimental results on long-video benchmarks show that EC-RAG consistently outperforms strong clip-based retrieval baselines and long-context video models, especially on questions requiring causal reasoning over large temporal gaps, while also achieving favorable efficiency in streaming settings. While EC-RAG systematically enhances objective reasoning accuracy, the framework exhibits the limitation. Due to the limited availability of existing benchmark resources, the applicability of the method to broader scenarios remains to be verified. In future work, we will focus on building evaluation datasets for ultra-long video scenarios.

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A Implementation Details

A.1 A. Algorithm Pseudocode

Algorithm 1: EC-RAG: Spatiotemporal Graph Construction and Retrieval Strategy

Input: Streaming video $\mathcal{V}$, Vision encoder Ψ_v , VLM, Embedder Ψ_e , Query Q , ASR Speech Islands $\mathcal{A}$, Thresholds $\tau_{event}, \tau_{bg}, \gamma, \tau_{dup}$, Patience $m = 2$

Output: Constructed EKG $\mathcal{G}$, Final answer A

// Phase 1: Perception Layer (Dual-Modal Sentinel Chunking)

```

1  $B \leftarrow \emptyset, C \leftarrow \emptyset$ ;
2 for each frame  $f_t \in \mathcal{V}$  do
3    $v_t \leftarrow \Psi_v(f_t); B \leftarrow B \cup \{v_t\}$ ;
4   if  $|B| \geq 2$  then
5      $D_{smooth}(t) \leftarrow \text{GaussianFilter}(1 - \cos(v_{t-1}, v_t))$ ;
6     if  $D_{smooth}(t) > \tau_{event}$  and is local optimum then
7        $t_{core} \leftarrow t$ ;
8        $t_{start} \leftarrow \text{TraverseBack}(t_{core}, \text{until } D_{smooth} < \tau_{bg} \text{ for } m \text{ consecutive frames})$ ;
9        $t_{end} \leftarrow \text{TraverseForward}(t_{core}, \text{until } D_{smooth} < \tau_{bg} \text{ for } m \text{ consecutive frames})$ ;
10      for each speech island  $I_{audio} = [t_{as}, t_{ae}] \in \mathcal{A}$  do
11        if  $I_{audio}$  intersects  $[t_{start}, t_{end}]$  then
12           $t_{start} \leftarrow \min(t_{start}, t_{as})$ ; // Extend left boundary if needed
13           $t_{end} \leftarrow \max(t_{end}, t_{ae})$ ; // Extend right boundary if needed
14       $C \leftarrow C \cup \{\text{Sample}(\mathcal{V}[t_{start} : t_{end}], 8.0 \text{ FPS})\}$ ;

// Phase 2: Cognition Layer (SES Abstraction & Graph Merging)
15  $\mathcal{G} \leftarrow \emptyset, S_{prev} \leftarrow \text{NULL}$ ;
16 for each chunk  $c \in C$  do
17    $SES_c \leftarrow \text{VLM.Generate}(c, \text{Prompt}=\text{"Extract SES"})$ ;
18    $\mathcal{G} \leftarrow \mathcal{G} \cup SES_c$ , Insert  $SES_c$  to VectorDB;
19   if  $S_{prev} \neq \text{NULL}$  then
20      $v_{post} \leftarrow \Psi_e(S_{prev}.\text{Post\_State})$ ;
21      $v_{pre} \leftarrow \Psi_e(SES_c.\text{Pre\_State})$ ;
22     if  $\cos(v_{post}, v_{pre}) > \gamma$  then
23       Create edge  $S_{prev} \xrightarrow{[:\text{TEMPORAL\_NEXT}]} SES_c$  in  $\mathcal{G}$ ;
24    $S_{prev} \leftarrow SES_c$ ;

// Phase 3: Retrieval & Generation (Hop Traversal & Deduplication)
25  $\text{Anchors} \leftarrow \text{VectorDB.Search}(\Psi_e(Q), \text{TopK} = 3)$ ;
26  $\mathcal{M}_{seen} \leftarrow \emptyset, Ctx \leftarrow \emptyset$ ;
27 for each anchor  $a \in \text{Anchors}$  do
28    $\text{SubGraph} \leftarrow \text{GraphDB.BFS\_Traversal}(a, \text{MaxHops} = 2)$ ;
29   for each node  $n \in \text{SubGraph}$  do
30     if  $\mathcal{M}_{seen} \neq \emptyset$  then
31        $S_{max} \leftarrow \max_{v \in \mathcal{M}_{seen}} \cos(\Psi_e(n.\text{Text}), v)$ ;
32     else
33        $S_{max} \leftarrow 0$ ;
34     if  $S_{max} \leq \tau_{dup}$  then
35        $Ctx \leftarrow Ctx \cup \{n.\text{Text}\}$ ;
36        $\mathcal{M}_{seen} \leftarrow \mathcal{M}_{seen} \cup \{\Psi_e(n.\text{Text})\}$ ;

37  $A \leftarrow \text{LLM.Generate}(Q, Ctx)$ ;
38 return  $A$ ;

```

B Prompt Engineering Details

To ensure the full reproducibility of the EC-RAG framework, we detail the exact prompt templates employed in both the State-Event-State (SES) graph construction and the final Dual-Store generation phases.

B.1 SES Abstraction Prompt (Perception & Cognition)

During the offline perception and cognition phases, a VLM is tasked with discretizing continuous video chunks into chronological causal graphs. To prevent entity confusion and action hallucination, we design strict instruction rules. Most notably, we introduce the “*Goldilocks Granularity*” rule to prevent actions from being either too vague or too kinematic, and the “*Visual Attribute Injection*” rule to strictly forbid the use of pronouns or generic IDs (e.g., forcing “the man in red” instead of “he” or “E1”). This ensures seamless downstream topological merging.

Prompt 1: SES Graph Construction

```
[SYSTEM PROMPT]
You are a state-of-the-art Video-to-Graph Parser designed for high-fidelity
Event-Causal Reasoning.
[Strict Graph Construction Rules]:
1. Task-Level Physical Actions (The 'Goldilocks' Granularity): Describe
specific, observable physical tasks and object interactions...
- DO NOT use vague umbrella terms (e.g., 'performing').
- DO NOT over-decompose into meaningless joint kinematics.
2. Visual Attribute Injection (CRITICAL): NEVER use generic IDs or pronouns.
You MUST refer to entities by distinct visual attributes (e.g., write 'The
man in the black t-shirt', NOT 'E1').
3. Micro-Detail Exhaustion: Capture secondary background events, holding
props, and screen text.
4. Direct Visual Evidence (CRITICAL): Preserve only directly visible
evidence. If text is blurry or uncertain, output an empty string.
5. Strict Causality: An 'Event' is a directed edge bridging a 'Pre-State'
and 'Post-State'.
6. Output Format: Output ONLY pure JSON.

[USER PROMPT TEMPLATE]
Target Timestamp: {start_time} - {end_time}.
{audio_context}
Task: Deconstruct this clip into a chronological causal graph.
Step 1: Inventory ALL interacting entities, detailing visual attributes.
Step 2: Map the specific, task-level physical actions.
Step 3: Preserve direct visual evidence useful for downstream QA.
Step 4: If any text is unreadable, write an empty string.
[Mandatory JSON Schema]: {scene_inventory: [...], events: [...]}
```

B.2 Dual-Store RAG Generation Prompt (Retrieval)

In the final QA generation phase, the LLM integrates both the original raw video frames and the retrieved multihop text subgraph (ctx). To mitigate potential hallucinations derived from the text graph, we explicitly establish a decision hierarchy (Rule 4), instructing the model to prioritize directly visible video evidence in the event of a conflict.

Prompt 2: Final QA Generation

You must answer a multiple-choice question about the attached original video. Choose exactly one option index from 0 to 4.

[Question & Choices]
{q}

[Decision Rules]

- 1) Use the original video frames as the primary evidence.
- 2) Use the retrieved graph memory as complementary evidence from the same video.
- 3) Compare all five options against the visible action, object, person, and purpose.
- 4) If video evidence and graph memory disagree, prefer the directly visible video evidence.
- 5) If evidence is incomplete, make the best forced choice from the available options.

[Retrieved RAG Memory]

The following graph memory was extracted from the same video and may help identify temporal actions, objects, participants, and causal context.

{ctx}

Output exactly one line and nothing else: [FINAL ANSWER: X]